higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

T760**(E)**(J28)T

AUGUST EXAMINATION

NATIONAL CERTIFICATE

INDUSTRIAL INSTRUMENTS N5

(8080205)

28 July 2015 (Y-Paper)
13:00–16:00

ENGINEERING
NATED

This question paper consists of 4 pages, 1 formula sheet and 5 graph sheets.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
INDUSTRIAL INSTRUMENTS N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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SECTION A

QUESTION 1

1.1 Prove, by means of Bernouli's theorem for liquids, that the velocity in a pipe, when measured by means of a pitot tube, is equal to:
$V = C\sqrt{2gh}$. (5)

1.2 Describe, with the aid of a sketch, the operating principle of a turbine flowmeter. (10)

1.3 Would you have to compensate for the square root function when measuring flow with a turbine meter? Motivate your answer. (2)
[17]

QUESTION 2

2.1 Calculate the bore diameter of an orifice plate if the following data is given:

- D + D/2 taps
- Maximum flow of gas is 738 m^3/h
- Maximum differential pressure is 1 000 Pa
- Diameter of the pipe is 25 cm
- Density of the gas is 6,408 kg/m^3
- The specific heat ratio is 1,4
- The pipe line pressure is controlled at 381 mm W.G
- Gas temperature is 20 ºC
- Viscosity is 1,28 x 10^{-4} poise (11)

2.2 What would the nett pressure loss across this orifice plate be? (2)
[13]

QUESTION 3

3.1 Calculate the flow of a liquid through a V-notch weir. The angle of the notch is 60º and height of the liquid is 60 cm above the crest of the V-notch.

$g = 10\ m/s^2$. (4)

3.2 Name FOUR types of weirs used. (4)
[8]

TOTAL SECTION A: 38

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SECTION B

QUESTION 4

4.1 What is understood by pH? (2)

4.2 What is the main purpose of a reference electrode? (3)

4.3 If the pH meter register an alkaline solution, would there be an abundance of *hydrogen ions* or *hydroxyl ions*? (1)

4.4 What would the pH of an alkaline solution register on a scale of 0–14? (1)

4.5 Why is screening and earthing important in pH measurement? (3)
[10]

QUESTION 5

5.1 Explain, with the aid of a neat sketch, the operation of the falling-piston viscometer. (5)

5.2 What is understood by Newtonian and non-Newtonian fluids? (3)
[8]

QUESTION 6

6.1 Explain, with the aid of a sketch, the operation of a dew-cell hygrometer. (8)

6.2 Make a neat, labelled sketch of a displacement type hygrometer. (5)
[13]

TOTAL SECTION B: 31

SECTION C

QUESTION 7

Discuss, with the aid of a principal sketch, the operation of a three-term electronic controller making use of one operational amplifier. **[16]**

QUESTION 8

8.1 Make a sketch of a closed-loop control system. (10)

8.2 Make a labelled sketch of a force-balance proportional action controller. (5)
[15]

TOTAL SECTION C: 31
GRAND TOTAL: 100

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INSUSTRIAL INSTRUMENTS N5

FORMULA SHEET

$W = 359{\cdot}2\ CZ\epsilon Ed^2\sqrt{(h\rho)}$ $\qquad R_d = W/15{\cdot}8\ \mu d$

$Q = 359{\cdot}2\ CZ\epsilon Ed^2\sqrt{(h/\rho)}$ $\qquad R_d = Q\rho/15{\cdot}8\ \mu d$

$W = 0{\cdot}01252\ CZ\epsilon Ed^2\sqrt{(h\rho)}$ $\qquad R_d = 3{\cdot}54\ W/\mu d$

$Q = 0{\cdot}01252\ CZ\epsilon Ed^2\sqrt{(h/\rho)}$ $\qquad R_d = 3{\cdot}54\ Q\rho/\mu d$

$Q_g = 2238\ CZ\epsilon Ed^2\sqrt{(h/\rho)}$ $\qquad R_d = Q_g\rho/98{\cdot}6\ \mu d$

$m = (d/D)^2$ $\qquad E = 1/\sqrt{(1 - m^2)}$

$$N = \frac{W}{0{\cdot}01252\ D^2\sqrt{(h\rho)}} = \frac{Q\sqrt{(\rho)}}{0{\cdot}01252\ D^2\sqrt{(h)}}$$

$mE = N/CZ\epsilon$ $\qquad CmE = N/Z\epsilon$ $\qquad mE = CmE/C$

$m = (d/D)^2$ $\qquad E = 1/\sqrt{(1 - m^2)}$

$$R_d = \frac{W}{15{\cdot}8\ \mu D\sqrt{(m)}} = \frac{Q\rho}{15{\cdot}8\ \mu D\sqrt{(m)}} = \frac{Q_g\rho}{98{\cdot}6\ \mu D\sqrt{(m)}}$$

$$N = \frac{W}{359{\cdot}2\ D^2\sqrt{(h)}} = \frac{Q\sqrt{(\rho)}}{359{\cdot}2\ D^2\sqrt{(h)}} = \frac{Q_g\sqrt{(\rho)}}{2238\ D^2\sqrt{(h}$$

$d/D = [(mE)^2/1 + (mE)^2]^{\frac{1}{4}}$ $\qquad mE = N/CZ\epsilon$

$W = 1890\ Ud^2\sqrt{(\rho P)}$ for critical flow/vir kritieke vloei

$CmE = N/Z\epsilon$ $\qquad mE = CmE/C$

$d = [W/1890\ U\sqrt{(\rho P)}]^{\frac{1}{2}}$ for critical flow/vir kritieke vloei

$$R_d = \frac{3{\cdot}54W}{\mu D\sqrt{(m)}} = \frac{3{\cdot}54Q\rho}{\mu D\sqrt{(m)}}$$ $\qquad d/D = [(mE)^2/1 + (mE)^2]^{\frac{1}{4}}$

$W = 1{\cdot}252\ Ud^2\sqrt{(\rho P}$ for critical flow/vir kritieke vloei

$d = [W/1{\cdot}252\ U\sqrt{(\rho P)}]^{\frac{1}{2}}$ for critical flow/vir kritieke vloei

1 kPa = 102 mmWD = 102 mmWG

1 lb/ft^3 = 16,0183 kg/m^3

Atmospheric pressure/Atmosferiese druk = 101,325 kPa

Gravitation acceleration/Gravitasieversnelling = 9,81 m/s^2

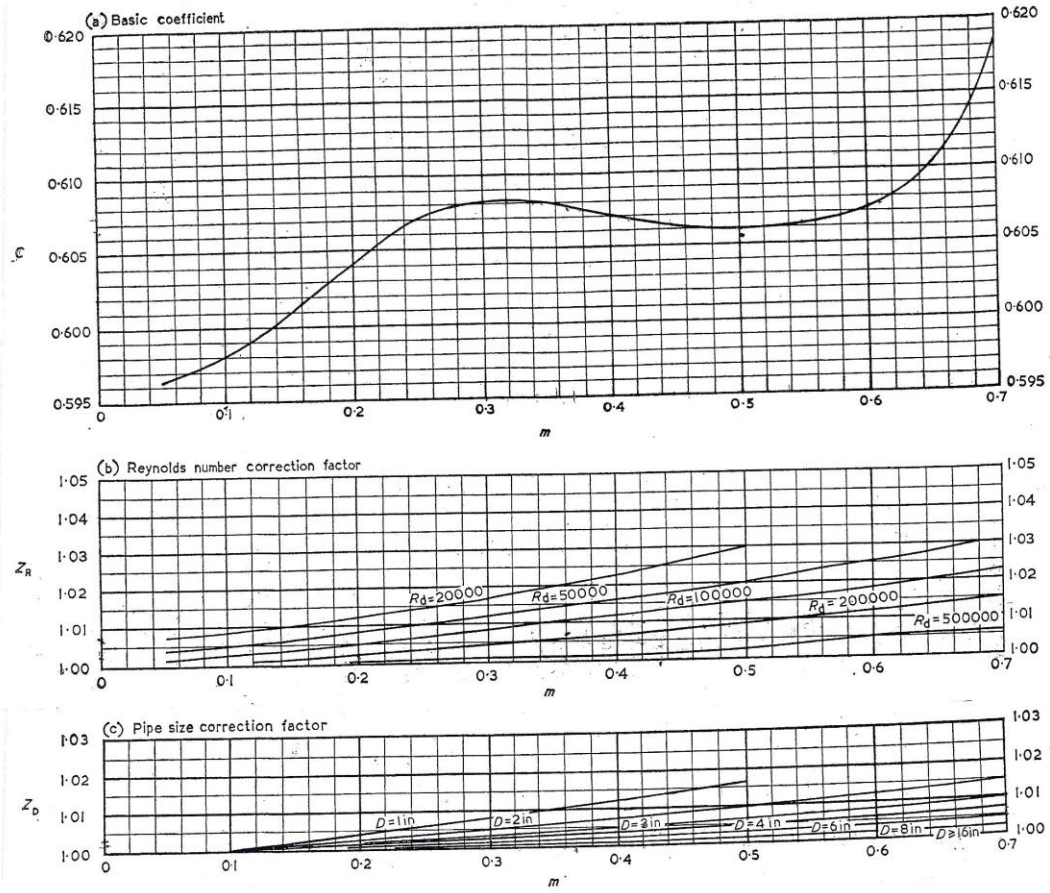
For/Vir D + $D/2$ tappings and/en flange tappings/flenstappe

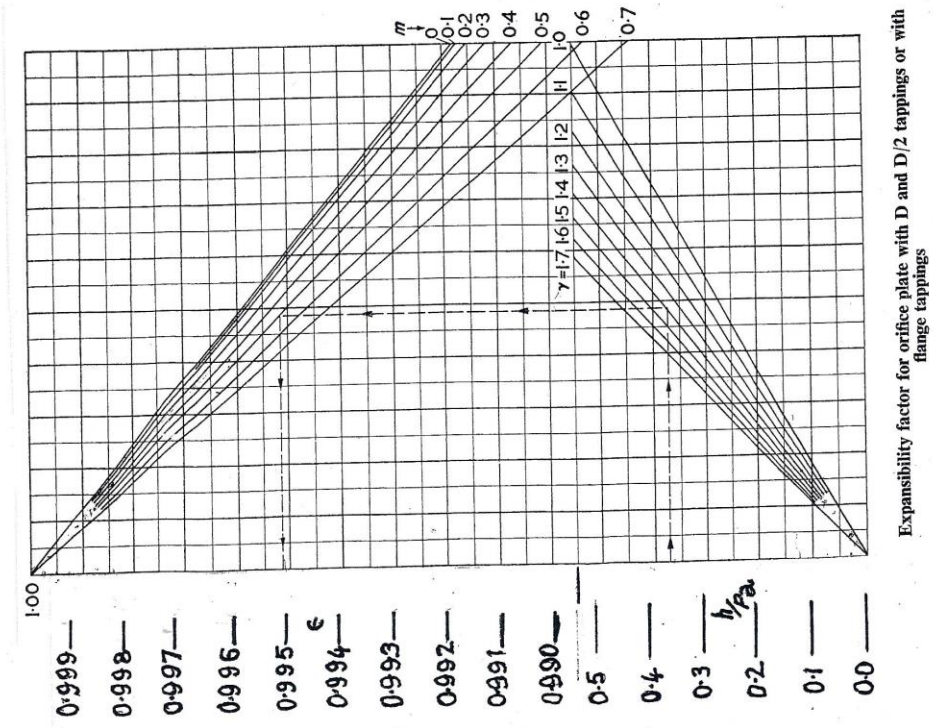
$$\frac{h}{Pa} \times 27{,}2 = \frac{kPa}{kPa} \times 27{,}2$$

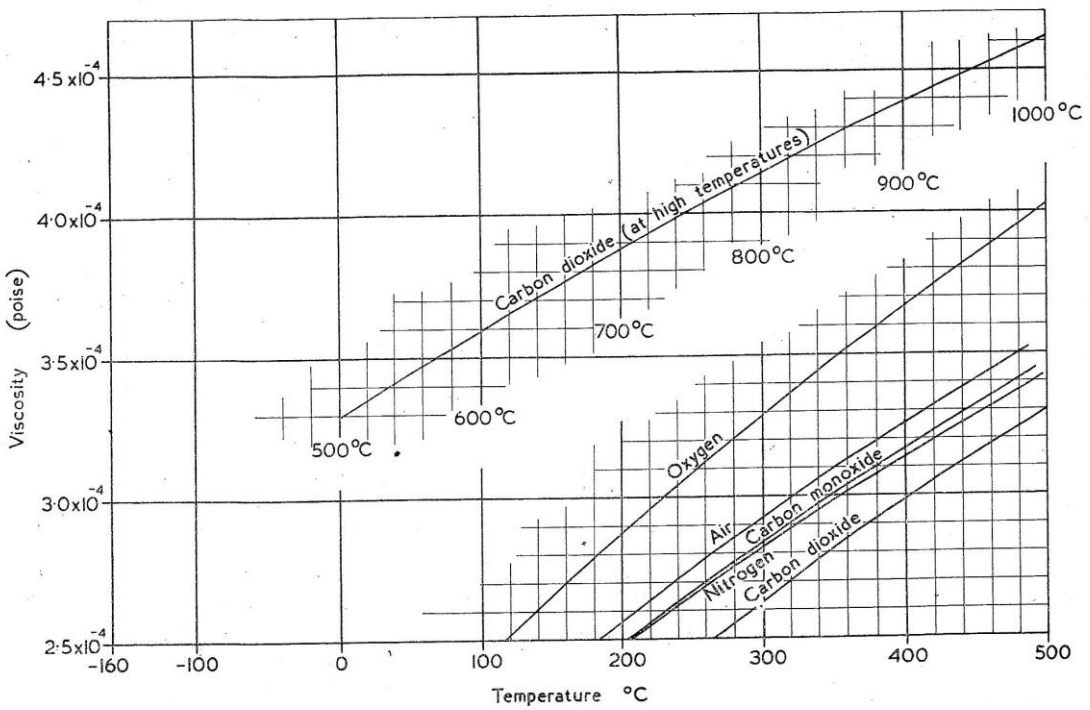
$$Q = \frac{8}{15}\ Tan\ \frac{\theta}{2}\ \sqrt{2g.H^5} \qquad Q = \frac{2}{3}\ B\ \sqrt{2g.H^3}$$

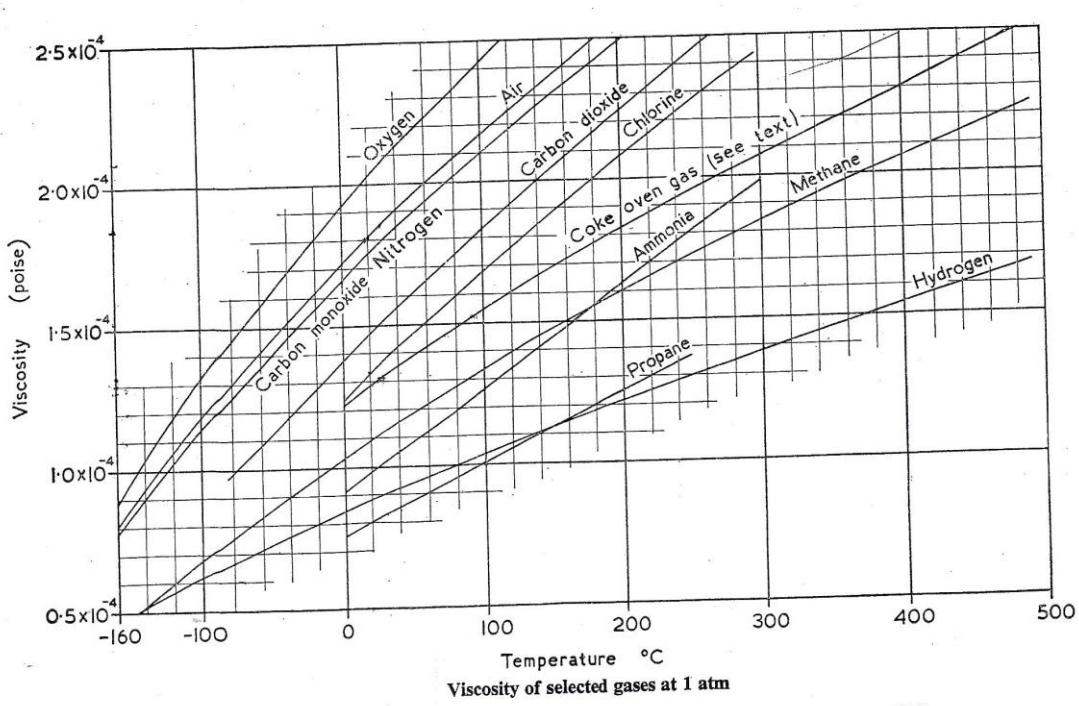
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GRAPH SHEET


(a) Basic coefficient
C
m
(b) Reynolds number correction factor
Z_R
R_d=20000
R_d=50000
R_d=100000
R_d=200000
R_d=500000
(c) Pipe size correction factor
Z_D
D=1in
D=2in
D=3in
D=4in
D=6in
D=8in
D≥16in


m
γ=1·7 1·6 1·5 1·4 1·3 1·2 1·1 1·0
1·00
ε
h/p₂
Expansibility factor for orifice plate with D and D/2 tappings or with flange tappings





Viscosity of selected gases at 1 atm

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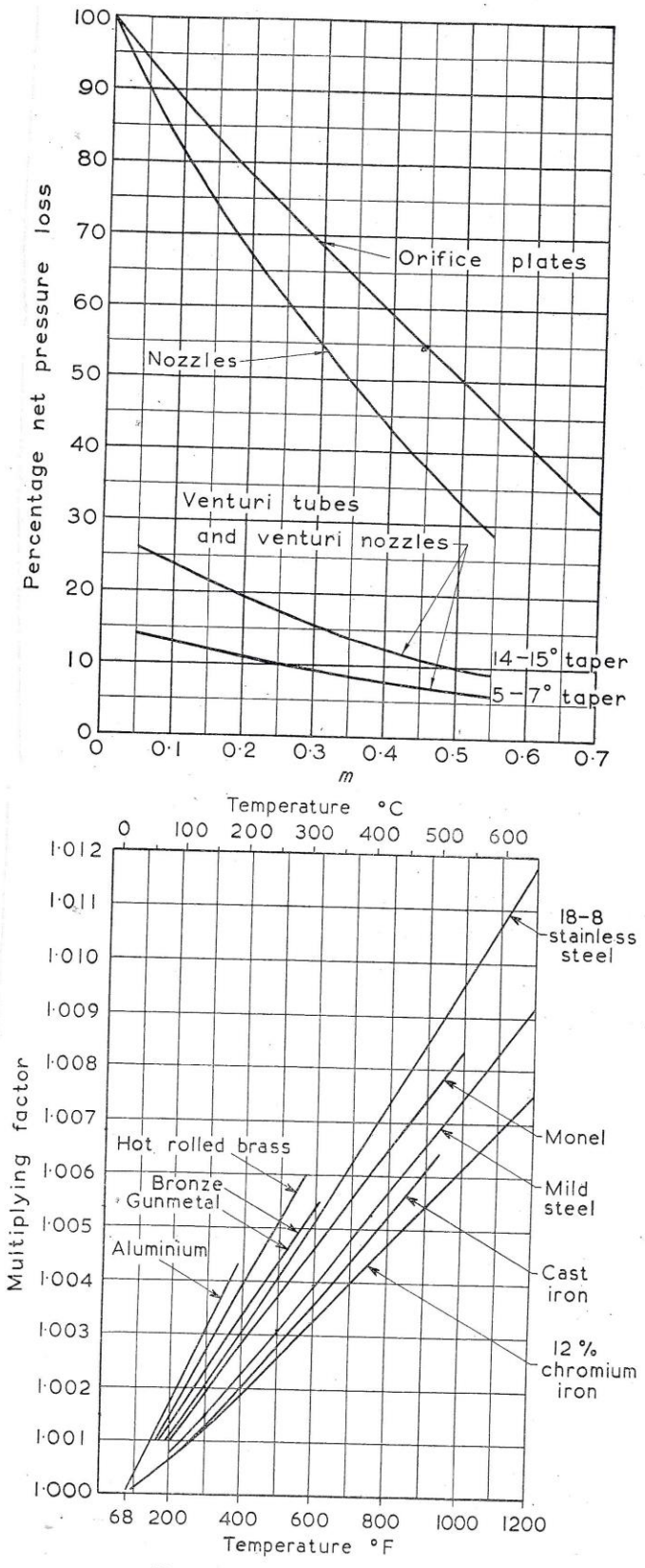
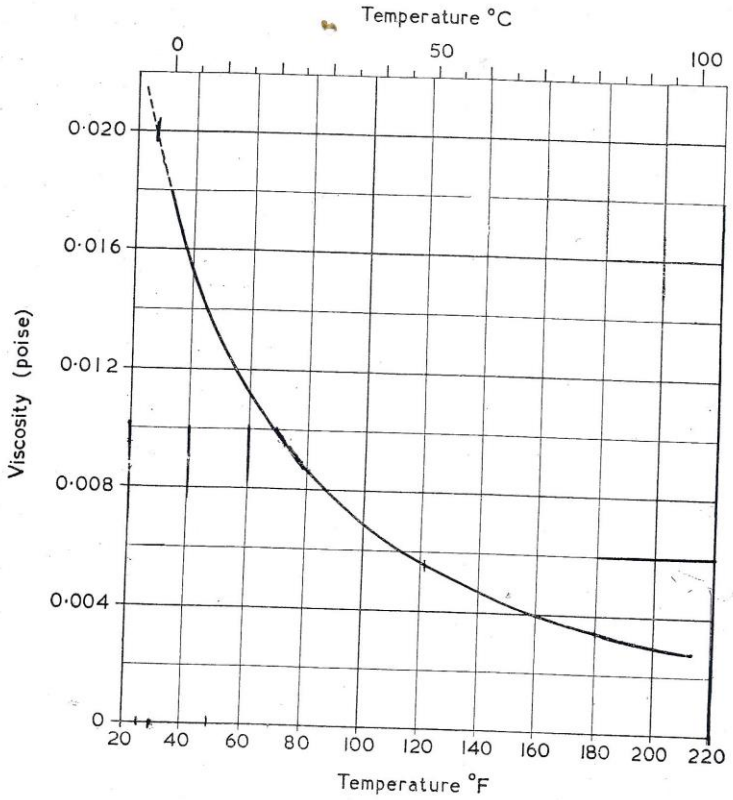
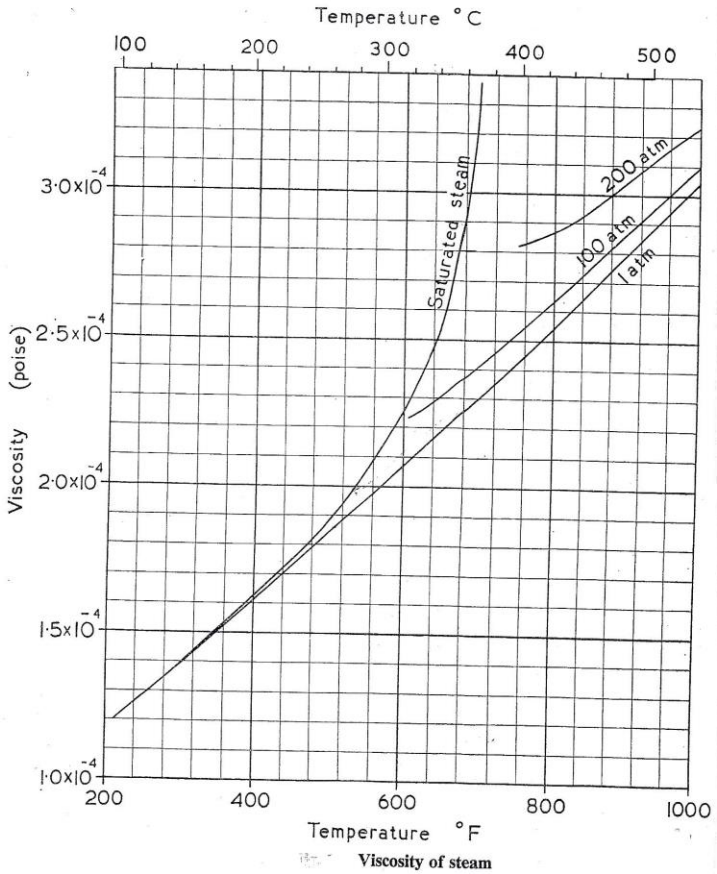


Fig. 5. Multiplying factors for thermal expansion



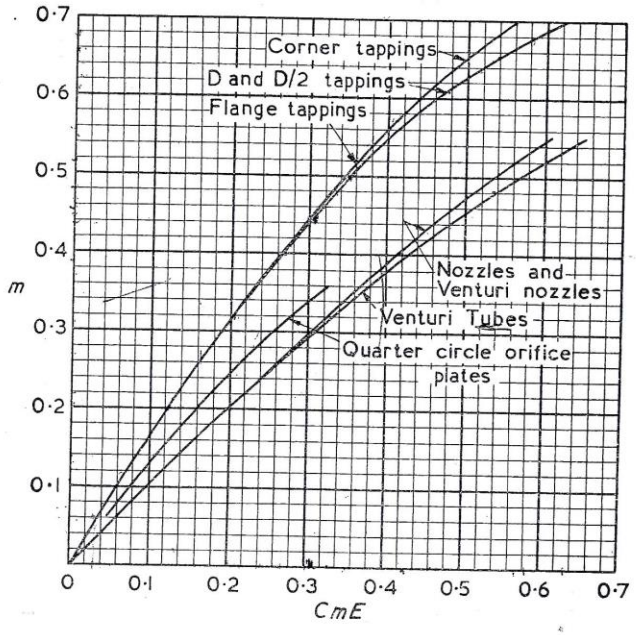
Viscosity of water at 1 atm



Viscosity of steam

TABLE 5. DENSITY (IN LB/FT³) OF SATURATED AND SUPERHEATED STEAM*

Absolute pressure (lbf/in²)	Saturated steam	Superheated steam at a total temperature of								
		250°F	300°F	350°F	400°F	450°F	500°F	550°F	600°F	650°F
14	0·03559	0·03351	0·03121	0·02922	0·02749	0·02595	0·02458	0·02335	0·02224	0·02123
15	0·03795	0·03593	0·03347	0·03133	0·02946	0·02781	0·02634	0·02502	0·02383	0·02275
16	0·04032	0·03834	0·03571	0·03342	0·03143	0·02966	0·02810	0·02669	0·02542	0·02427
17	0·04266	0·04080	0·03798	0·03555	0·03342	0·03154	0·02987	0·02837	0·02702	0·02579
18	0·04500	0·04322	0·04023	0·03765	0·03539	0·03339	0·03162	0·03004	0·02860	0·02731
19	0·04735	0·04566	0·04248	0·03976	0·03737	0·03526	0·03340	0·03172	0·03019	0·02883
20	0·04968	0·04810	0·04474	0·04186	0·03934	0·03713	0·03516	0·03339	0·03180	0·03034
22	0·05432	0·05299	0·04926	0·04608	0·04329	0·04085	0·03868	0·03674	0·03498	0·03338
24	0·05893	0·05790	0·05382	0·05033	0·04726	0·04458	0·04221	0·04008	0·03815	0·0364^
26	0·06353	0·06281	0·05838	0·05456	0·05123	0·04833	0·04575	0·04342	0·04136	0·0394[illegible]
28	0·06812	0·06775	0·06289	0·05879	0·05522	0·05206	0·04929	0·04679	0·04456	0·0425[illegible]
30	0·07273	0·07267	0·06748	0·06305	0·05917	0·05580	0·05283	0·05015	0·04776	0·04556
32	0·07722	—	0·07215	0·06739	0·06321	0·05959	0·05640	0·05353	0·05097	0·04861
34	0·08177	—	0·07675	0·07163	0·06720	0·06337	0·05995	0·05691	0·05417	0·05168
36	0·08621	—	0·08130	0·07587	0·07117	0·06711	0·06349	0·06024	0·05737	0·05473
38	0·09066	—	0·08591	0·08019	0·07519	0·07087	0·06702	0·06361	0·06057	0·05777
40	0·09515	—	0·09058	0·08446	0·07924	0·07468	0·07062	0·06698	0·06378	0·06083
42	0·09970	—	0·09524	0·08881	0·08326	0·07843	0·07413	0·07037	0·06698	0·06390
44	0·1041	—	0·09990	0·09311	0·08726	0·08217	0·07770	0·07375	0·07022	0·06693
46	0·1086	—	0·1045	0·09737	0·09132	0·08598	0·08137	0·07716	0·07342	0·07003
48	0·1130	—	0·1093	0·1017	0·09542	0·08985	0·08489	0·08052	0·07663	0·07305
50	0·1174	—	0·1140	0·1061	0·09950	0·09363	0·08849	0·08389	0·07987	0·07616
52	0·1218	—	0·1187	0·1104	0·1035	0·09737	0·09208	0·08734	0·08313	0·07924
54	0·1262	—	0·1234	0·1148	0·1075	0·1012	0·09560	0·09074	0·08636	0·08230
56	0·1305	—	0·1281	0·1192	0·1116	0·1050	0·09921	0·09416	0·08952	0·08540
58	0·1349	—	0·1329	0·1236	0·1157	0·1089	0·1029	0·09756	0·09276	0·08842
60	0·1393	—	0·1377	0·1280	0·1198	0·1127	0·1065	0·1009	0·09597	0·09149



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